

Generation of Amorphous Silicon Models via Melt-Quench Molecular Dynamics with Moment Tensor Potentials

William Homier¹,
Supervisor: Prof. Lena Simine², and Co-mentor: Ata Madanchi³

¹*McGill University Chemistry, 801 Sherbrooke St. West, Montréal, QC H3A 2T8,
Canada*

July 27th, 2026

Abstract

1 Introduction

1.1 Objectives

2 Background and Methods

2.1 Molecular Dynamics and the Melt-Quench Protocol

2.2 Interatomic Potentials: From DFT to MTP

2.3 Computational Setup

3 Preliminary Results

Figure 1: Placeholder: radial distribution function of quenched a-Si.

4 Discussion

5 Remaining Work and Timeline

6 Conclusion

Acknowledgements

References